

PLANRS ACADEMY MASTER COURSE

Grade 4 | Graduation Showcase

GRADUATION SHOWCASE

The Junior Innovator's Guide to Design Thinking

Welcome to the World of Design!

Did you know that you are already a designer? Every time you find a way to make something work better, you are using Design Thinking.

What is a Gadget?

A gadget is more than just a piece of technology. It is a tool that solves a problem for people. It can be as simple as a tool to pump air or as complex as a computer.

Your Mission Today:

In this lesson, you will learn how real heroes use simple materials to solve big problems, and you will start your own journey using the Design Loop.

Mini-Task: Look around right now. Spot 3 gadgets you use every day. Think about your water bottle, cycle pump, or lunchbox.

1. Functional Design

Design is not just about how things look—it's about how they work! We call this **Functional Design**: design made to serve a special purpose.

- **The Idli Stand:** A brilliant design that saves space by stacking plates on top of each other.

- **The Cycle Pump:** A functional tool that keeps your bike ready to ride whenever you need it.

2. Meet Real-Life Heroes

Meet two amazing Indian innovators who turned simple ideas into gadgets that changed lives:

Mansukhbhai Prajapati (The Clay Innovator):

He created *Mitticool*, a refrigerator made completely out of clay! It keeps water, milk, and vegetables fresh and cool without using any electricity.

Remya Jose (The Bicycle Laundry Star):

When she was a school student, she invented a washing machine attached to a bicycle! By pedalling the cycle, she could wash clothes while exercising, saving both electricity and time.

3. The Design Loop: THINK

How do inventors come up with great ideas? They use a map called the **Design Loop**! The loop has three main steps that keep going around: THINK, MAKE, and SHARE. Let's look at the first step.

STEP 1: THINK - Find the Problem

Every great gadget starts by looking around and finding a problem to solve. Design thinkers are always asking questions like:

-  What is hard to do?
-  What breaks often?
-  How can we make this task more fun or faster?

Brainstorming Challenge:

Look at this list of daily hassles. Which one would you like to build a gadget for?

- The water bottle is too heavy to carry.
- Pencils keep rolling off the desk and losing their tips.
- The tiffin box leaks oil into the school bag.

4. The Design Loop: MAKE

Once you have an idea, it's time to bring it to life!

STEP 2: MAKE - Build and Test

This is where you become a maker! You use simple materials to create a **Prototype**—a rough model of your gadget.

The Materials Station:

You don't need expensive factory tools. You can make an amazing prototype using things found around your house or classroom:

 Cardboard boxes, plastic cups, and bottles

 Rubber bands, popsicle sticks, and straws

 Paper clips, string, tape, and clay

[Visual Idea: A blueprint drawing setup of cardboard pieces and rubber bands showing what your prototype might look like.]

5. The Design Loop: SHARE

An invention isn't finished until you share it with others!

STEP 3: SHARE - Tell Your Story

Show your gadget to your friends, family, and classmates. Tell them how it helps solve the problem you found.

Why Sharing Matters:

-  **Get Better Ideas:** When others see your gadget, they might have great suggestions to improve it.
-  **Inspire Others:** Your invention could inspire someone else to create something amazing too!

Sharing Challenge:

Prepare a "30-second pitch" for your gadget. Explain:

1. What the problem was.
2. How your gadget solves it.
3. One thing you are proud of!

6. 3-2-1 Exit Review

Let's check what we've learned today to make sure you're ready to be an inventor!

3 Gadgets:

List 3 gadgets you learned about today (from the lesson or your surroundings).

2 Steps:

Write down 2 steps of the Design Loop that you remember.

1 Feeling Word:

What is one word that describes how you feel about being an innovator?

Your Final Challenge!

Find one problem at home or school.

Use the Design Loop to **THINK**, **MAKE**, and **SHARE** your gadget!

Bring your idea or prototype to the graduation showcase showcase.



Future Innovator Ready!